



FIT3080 – Artificial Intelligence

Introduction to Machine Learning Chapters 19.1-19.4.1, 19.6, 19.7.1

Learning Objectives (I)

- **Machine learning – definition**
 - Issues about learning from data
 - Concepts
- **Approaches to learning from data**
 - Verification driven
 - Discovery driven – inductive learning
- **Mathematical principles**
- **Supervised machine learning**
 - Decision Trees



Learning Objectives (II)

- **Supervised machine learning**
 - Regression
 - Linear classifiers with thresholds
 - Logistic regression
 - Artificial Neural Networks
 - Naïve Bayes
 - k Nearest Neighbour (k-NN)
- **Unsupervised learning**
 - Clustering
 - > Similarity measures
 - > The K-means algorithm



Assumptions about the Environment

- **Fully/partially observable**
- **Unknown**
- **Single/multi agent**
- **Deterministic/stochastic**
- **Sequential/episodic**
- **Static**
- **Discrete/continuous**



Types of Learning Approaches

- **Verification Driven**

- An expert formulates a hypothesis, which is validated by analyzing the data

- **Discovery Driven – Machine Learning**

- Programs learn from data
- Hypotheses are formulated bottom-up from the data
 - > Interesting patterns in the data are noticed and evaluated by the user/expert
- Machine learning performs ***inductive learning***, i.e., learning from examples



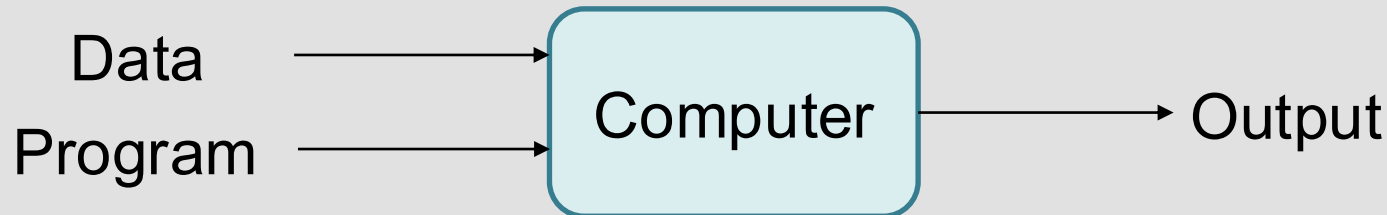
Overview of Machine Learning

- **Definition**

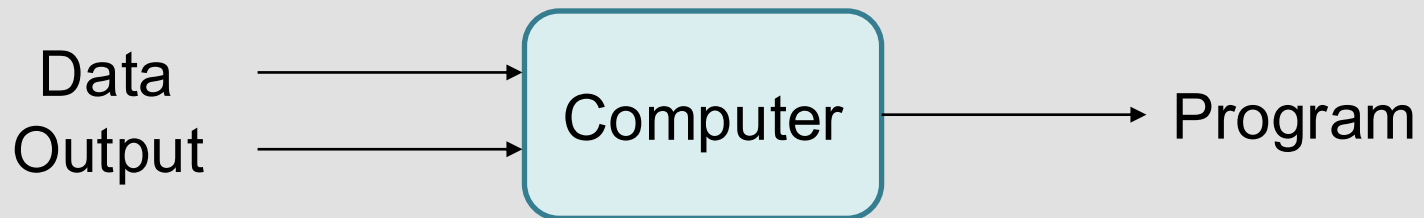
“Machine Learning is a field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel (1959)

Traditional Programming:



Machine Learning:



Why Machine Learning?

- **Machine Learning is used when**
 - Environments are unknown
 - Environments are dynamic
 - People can't explain their expertise
 - Models must be customized to specific situations
 - Models are based on large amounts of data
- **Learning modifies an agent's decision mechanisms to improve performance**



How are things learned?

- **Memorization**

- Accumulation of individual facts
- Limited by
 - Time
 - Memory

- **Generalization**

- Inducing new facts from old facts
- Predictive activity
 - Assumes that the past predicts the future
- Limited by the accuracy of the induction process





FIT3080 – Artificial Intelligence

Machine Learning Concepts

Representation – Learning Concepts

- **Machine learning aims to reveal knowledge about data**
 - concepts, definitions, hypotheses
- **A learning algorithm considers many instances and infers useful regularities about them**
 - It can ***infer (learn)*** a definition of a class
 - > this definition may be viewed as ***patterns*** or ***structures*** within the data
- **The structure of the discovered patterns depends on the learning techniques**
 - E.g., rules, decision trees



Evaluation – Completeness and Consistency

- **A definition is**

- **complete** if it recognizes all instances of a class
- **consistent** if it does not classify any negative examples as instances of this class

- **An *incomplete* definition is too narrow**

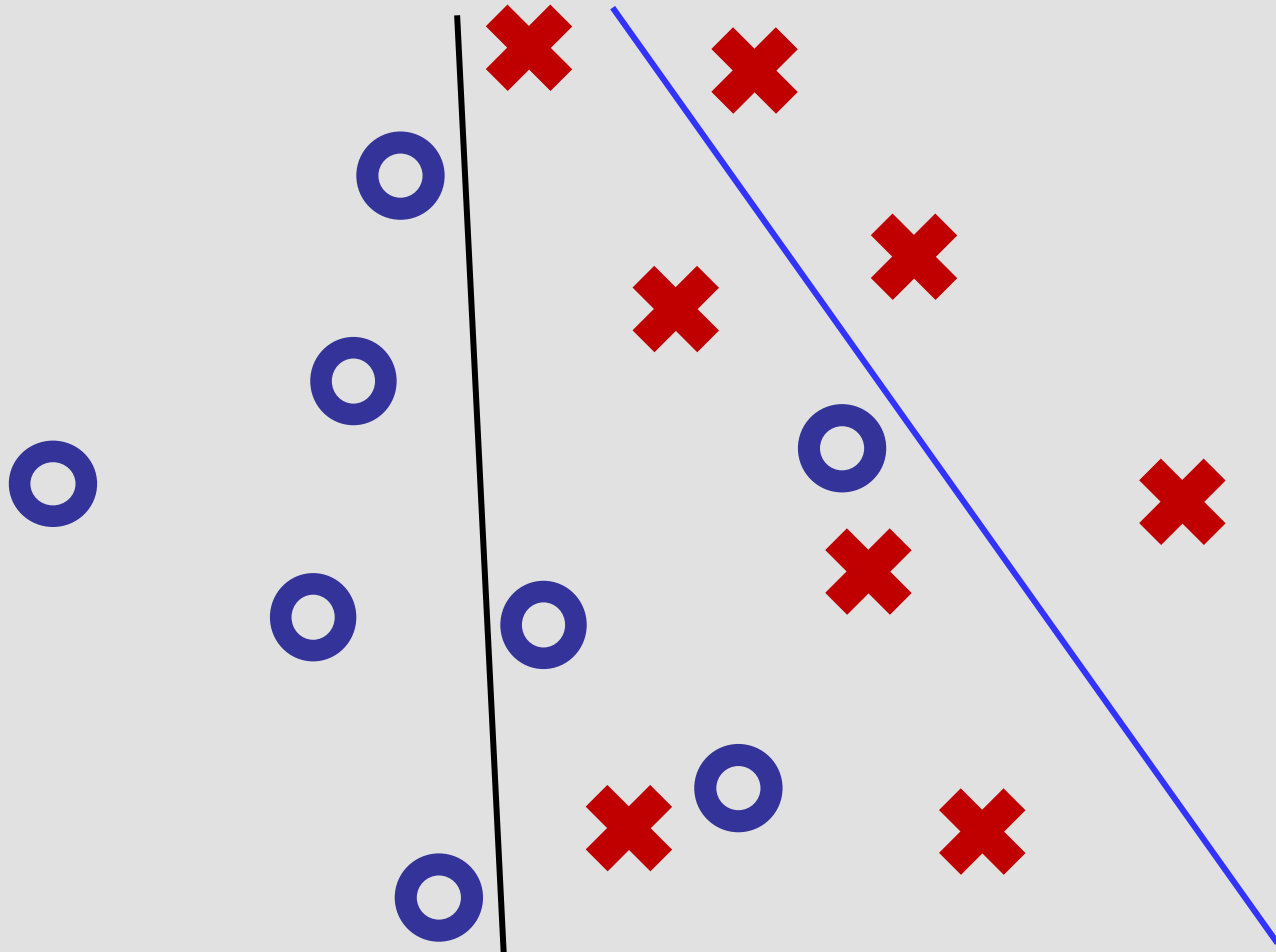
- it would **not** recognize some “in-class” instances as belonging to the class

- **An *inconsistent* definition is too broad**

- it would classify some “out-of-class” instances as belonging to the class



Example – Separate two classes



Machine Learning as Search

- **Find the best hypothesis among many**
 - Need a measure of the quality of candidate hypotheses
- **The search space is very large**
- **We don't know the optimal solution**



Machine Learning as Compression

- **We have learnt something if we have created a description of the data that is shorter than the original data set**
- **Example – Input/Output**
 - **Input:** a table of instances
 - **Output:** a prediction of an instance given a set of features or a suggested clustering of the instances



Machine Learning Principles

- ***“A general law can never be verified by a finite number of observations. It can, however, be falsified by only one observation.”***

Karl Popper

- ***“All models are wrong, but some models are useful.”***

George Box

- **Any results discovered *must* be tested for performance and statistical significance**



Types of Machine Learning

- *Supervised learning*
- *Unsupervised learning*
- *Reinforcement learning*: occasional rewards (or punishments) are given
 - E.g., Q-learning



Supervised Machine Learning

- **Build patterns to infer an unknown attribute given the values of known attributes**
- **Supervision:**
 - a set of observations (***training data***) is accompanied by labels indicating the class of each observation
 - new data are classified/predicted based on the model learned from the training data
- **Examples of supervised techniques:**
 - Decision Trees, Naïve Bayes, K-Nearest Neighbour (k-NN), Regression, Neural Nets



Unsupervised Machine Learning

- **Return “interesting” patterns in the data**
- **Lack of supervision:**
 - given a set of observations (*training data*), infer patterns in the data
 - training data is unlabelled – there are no pre-defined classes
- **Examples of unsupervised techniques :**
 - E.g., K-Means, Snob (Minimum Message Length Principle)



General Unsupervised Techniques

- **Clustering –**

- Groups records of similar items
 - The user must attach meaning to the clusters formed
- Example: Identify different types of customers

- **Association analysis –**

- Discovers relations hidden in the data
 - > represented in the form of ***association rules*** or ***sets of frequent items***
- Example: Market basket analysis

- **Outlier detection –**

- Discovers exceptions to normal patterns in the data
- Example: Intrusion detection





FIT3080 – Artificial Intelligence

Supervised Machine Learning

Supervised Learning – Principal Techniques

- **Classification – infers categorical or discrete values**
- **Regression – infers continuous or ordered values**
 - Logistic regression infers discrete values



Classification – Example (I)



Cat

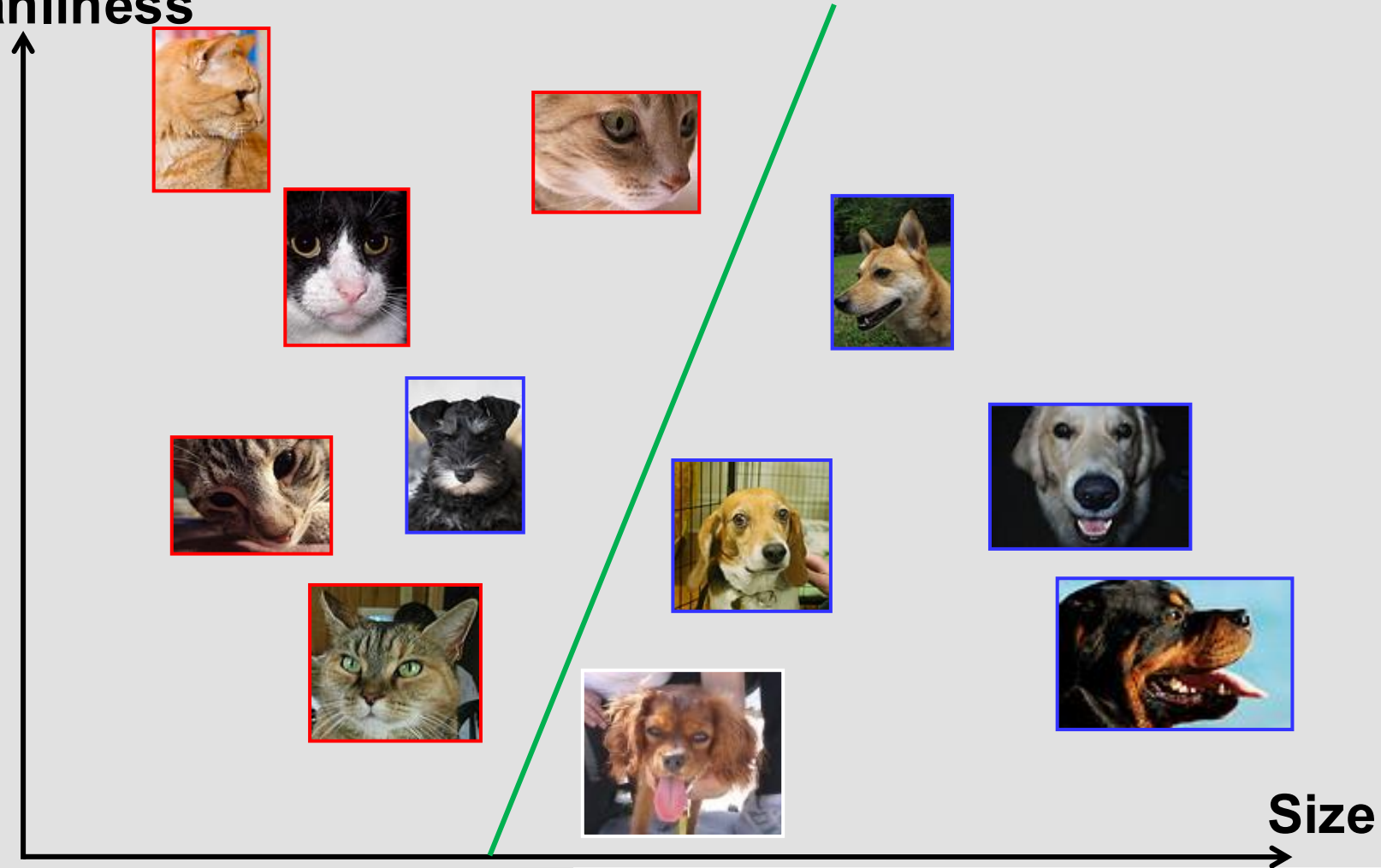
?

Dog



Classification – Example (II)

Cleanliness



Regression – Example (I)

\$

\$\$

\$\$\$

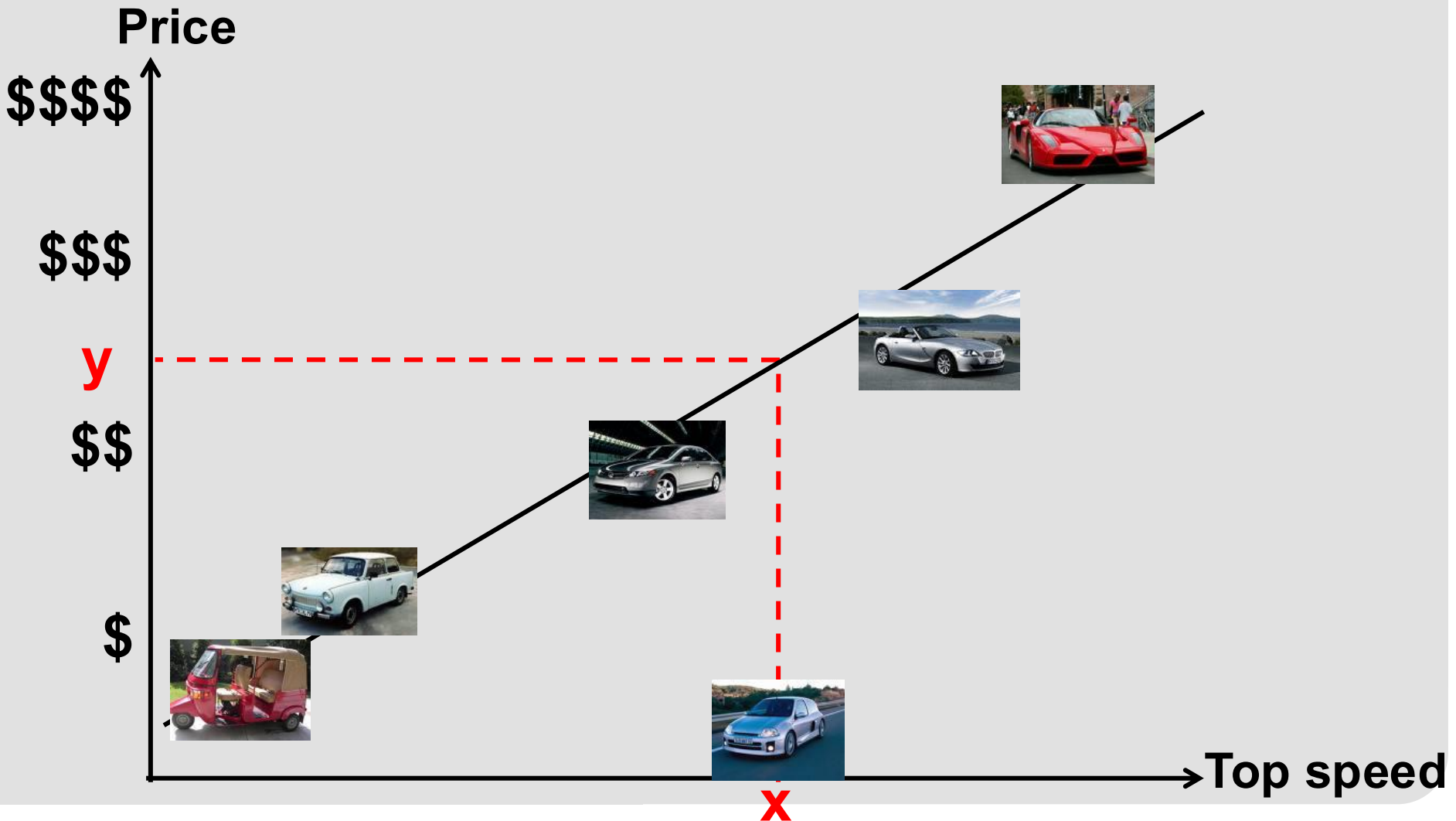
\$\$\$\$



?



Regression – Example (II)



Time of Inferred Value – Classification vs Prediction

- **Classification**

- Considers the features of an object in order to assign it to a pre-defined class
- Example: determining the species of an animal

- **Prediction**

- Classification of future events – the model is applied to historical data to predict future outputs
- Example: predict whether a share price will go up or down





FIT3080 – Artificial Intelligence

Supervised Machine Learning Process

Supervised Learning Process (I)

1. Model construction

- Build a model using a **Training Set**

2. Model evaluation – estimates the *performance* of the model

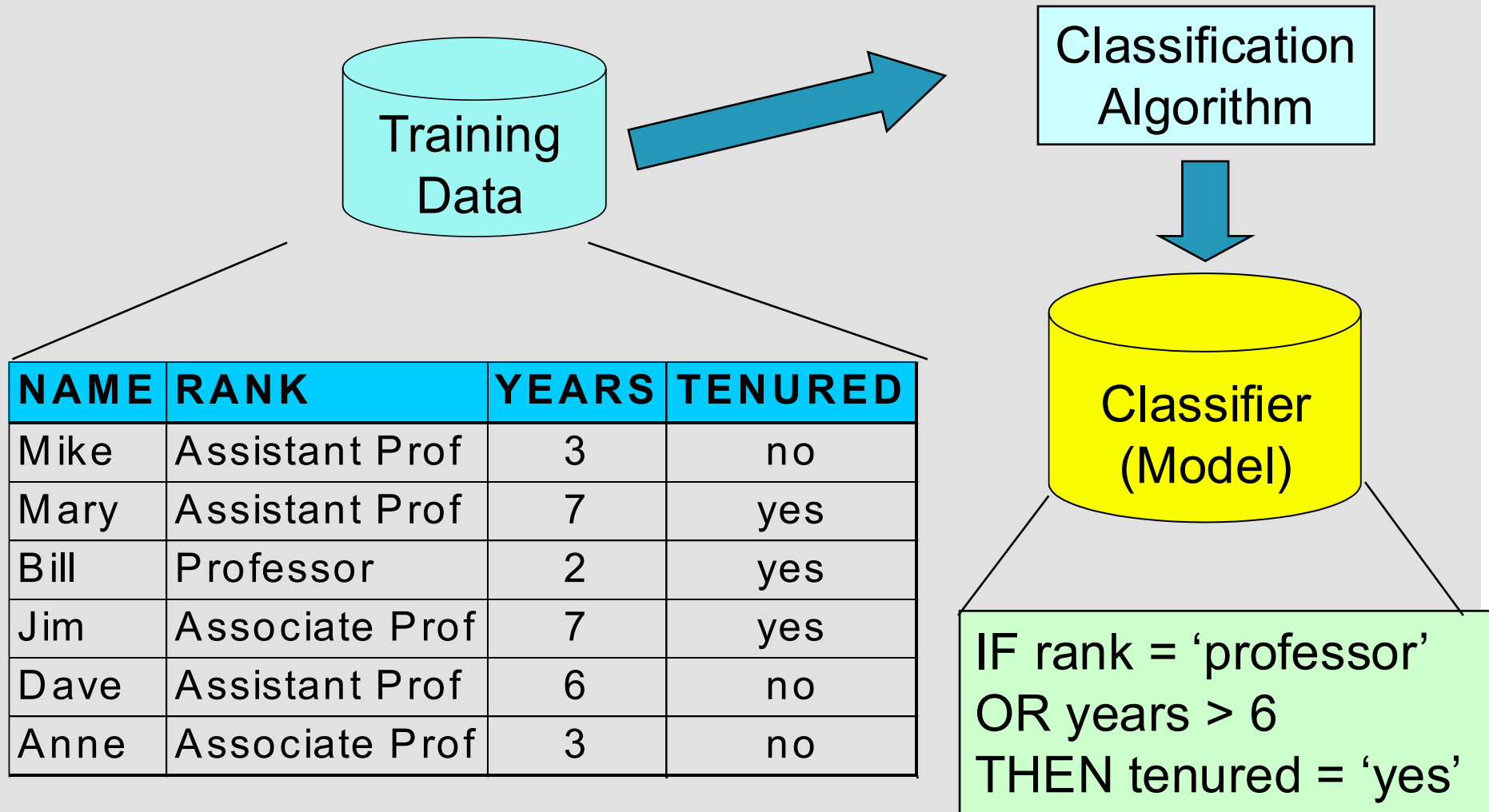
- Use a **Test Set**, which is different from the training set

3. Model usage

- Apply the model to unseen items



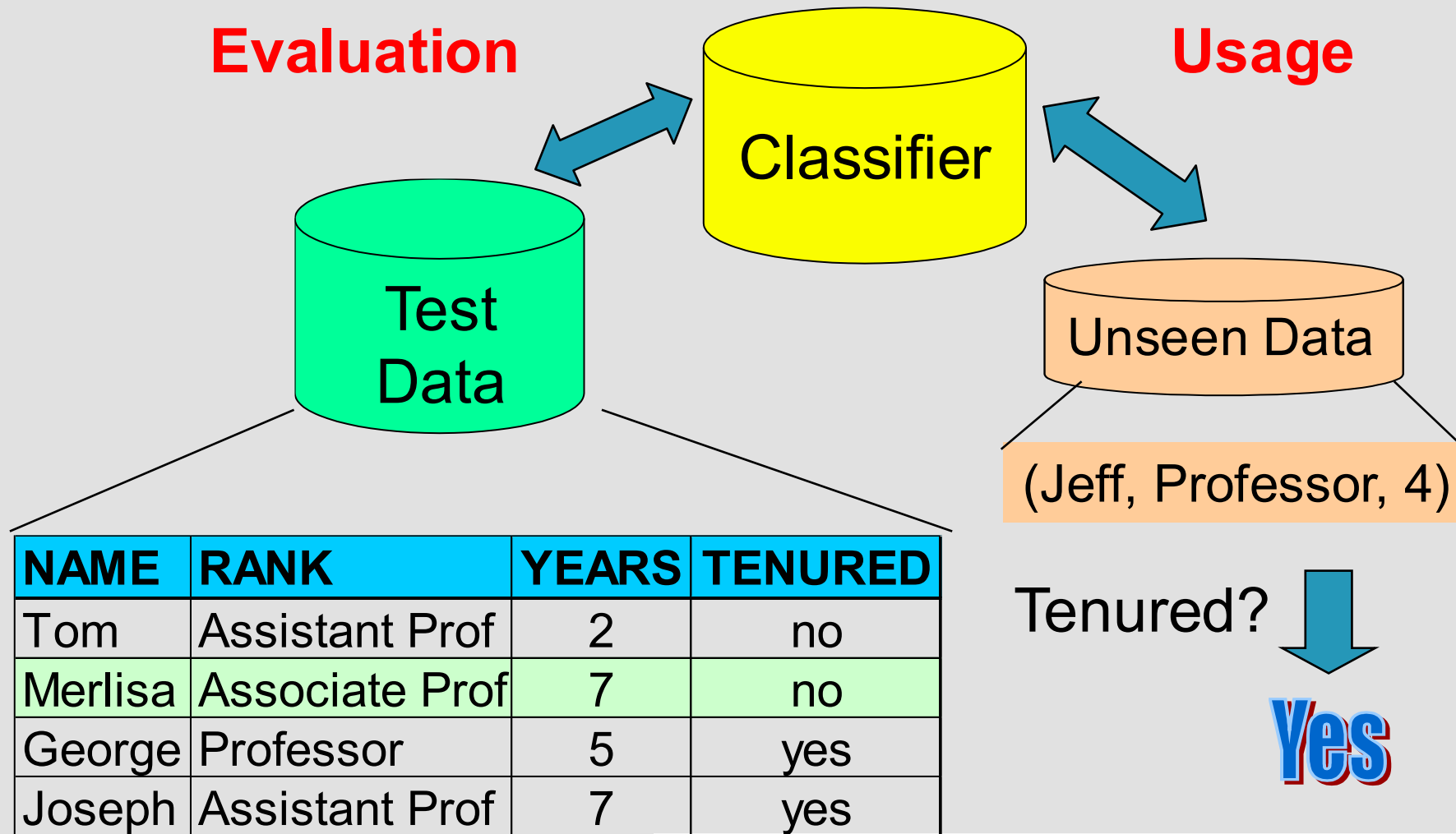
Classification – Model Construction



Classification – Model Evaluation and Usage

Evaluation

Usage



Supervised Learning Process (II)

	Classification	Regression
Model construction	<features, <i>class</i> >	<features, <i>value</i> >
Model evaluation	<i>class</i> vs predicted	<i>value</i> vs predicted
Evaluation metrics	Accuracy, recall, precision	RMSE (root mean squared error)
Model usage	Predict class of unseen items	Predict value of unseen items



Criteria for Evaluating ML Methods (I)

- **Hypothesis**
 - **Performance**
 - > Different measures, e.g., accuracy, recall, precision
 - **Significance**
 - > Must perform significantly better than the naïve hypothesis
 - **Model size** (related to information content)
 - > E.g., decision tree size, compactness of classification rules
 - **Transparency** (interpretability)
 - > Ease of understanding the model



Criteria for Evaluating ML Methods (II)

- **Practicalities of the model**
 - **Robustness**
 - > Ability to handle noise and missing values
 - **Scalability**
 - > Speed – time to build the model, and time to use the model
 - > Size – effect of the size of the dataset on the model





FIT3080 – Artificial Intelligence

Supervised Machine Learning – Training and Testing

Training, Validation and Test Data

- **Data: labeled instances**

Used to train the model

Training
Data

Used to fine-tune the model
and learn hyper-parameters

Validation
Data

Used to measure the
generalization of the model

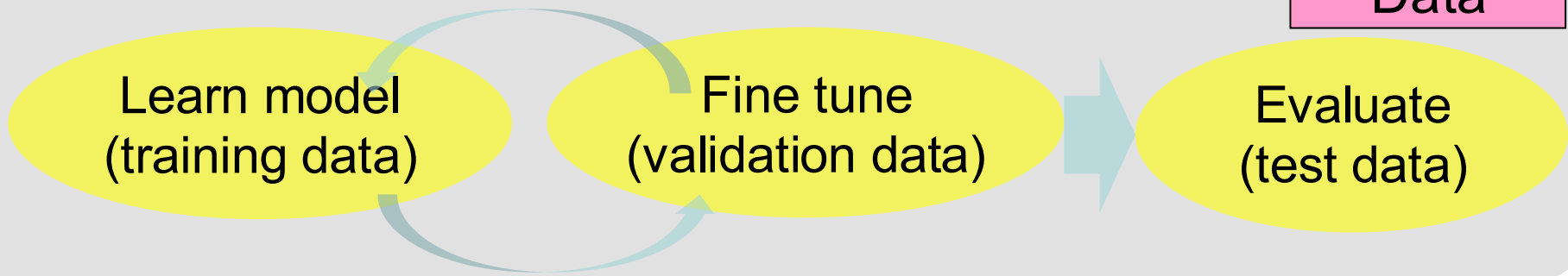
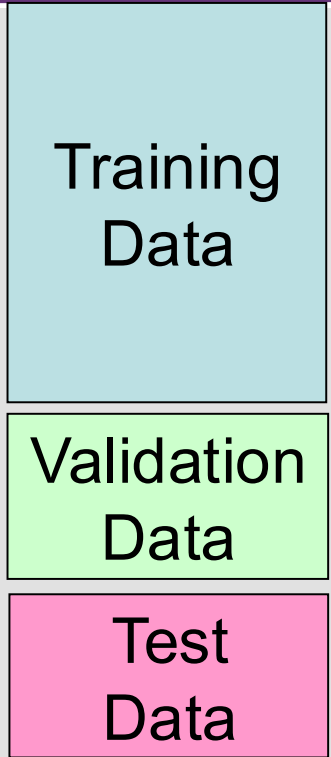
Test
Data



Supervised Learning Process (III)

- **Features: attribute-value pairs which characterize each instance**
- **Experimentation cycle**
 - **Learn model** parameters on **Training Data**
 - **Fine tune the model on Validation (Held-out) Data**
 - **Compute performance** on **Test Data**

Very important: never “peek” at the test set!



Classification Evaluation Metrics (I)

- accuracy $\frac{\text{correctly predicted}}{\text{predicted}}$

If 80 predictions are correct out of 100 then accuracy = $80/100 = 0.8$

- recall $\frac{\text{correctly predicted as class } C}{\text{instances in class } C}$
- precision $\frac{\text{correctly predicted as class } C}{\text{predicted as class } C}$
- F-score $\frac{2 \text{ Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$

Training
Data

Validation
Data

Test
Data



Classification Evaluation Metrics – Example

Who will get HD in FIT3080?

Calculate accuracy, recall and precision for Model A

	HD	Not HD
Ground truth	A, B, C, D, E, F	G, H, I, J, K, L, M, N, O
Model A	A, B, C, D	E, F, G, H, I, J, K, L, M, N, O

Model A

$$\bullet \text{ Accuracy} = \frac{\text{correct HD} + \text{correct NotHD}}{\text{all predictions}} = \frac{4 + 9}{15} = \frac{13}{15}$$

$$\bullet \text{ Recall HD} = \frac{4}{6} = 0.667 \qquad \text{Recall NotHD} = \frac{9}{9} = 1$$

$$\bullet \text{ Precision HD} = \frac{4}{4} = 1 \qquad \text{Precision NotHD} = \frac{9}{11}$$



Classification Evaluation Metrics (II)

- **Micro averaging: gives equal weight to every instance**

– Example:

$$accuracy = \frac{correct_A + correct_B + correct_C}{all\ predictions}$$

correctly
predicted

- **Macro averaging: gives equal weight to every class**

– Example:

$$accuracy = \frac{1}{3} \left(\frac{correct_A}{predict_A} + \frac{correct_B}{predict_B} + \frac{correct_C}{predict_C} \right)$$



Evaluation – Cross-validation

- **Performance**
 - depends on the representativeness of the training data
 - determined using a **test set**
- **Need to perform cross-validation, Why?**
 - **X-validation** – repeated runs on different training/test sets
 - > separate the dataset into several training and test sets
 - 10-fold X-validation: repeated 90/10% train/test
 - 5-fold X-validation: repeated 80/20% train/test
 - leave-one-out
 - > build a model from the training set, compute performance on the test set
 - **Stratified X-validation** – the test sets are proportional to the classes in the data
 - > e.g., 20% +ive, 80% -ive



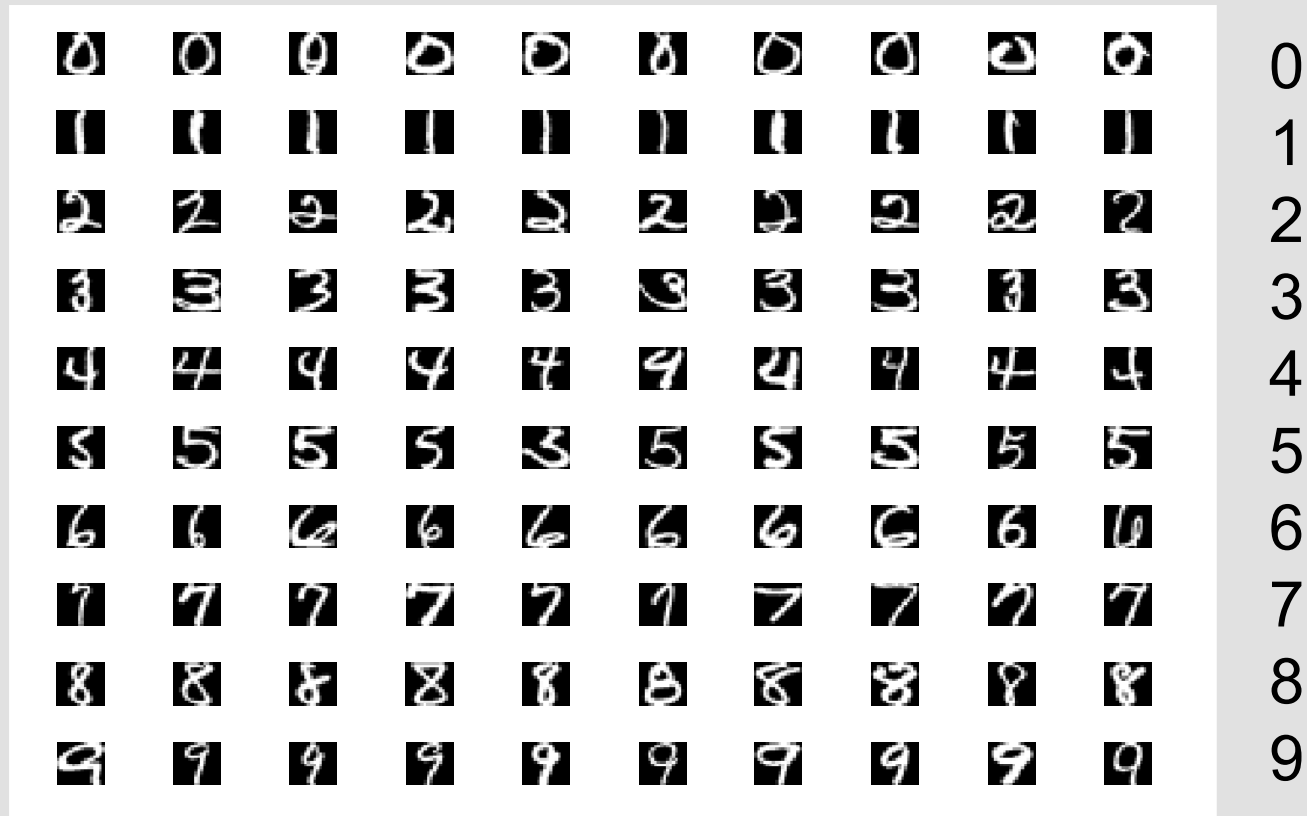


FIT3080 – Artificial Intelligence

Supervised Machine Learning – Mathematical Principles

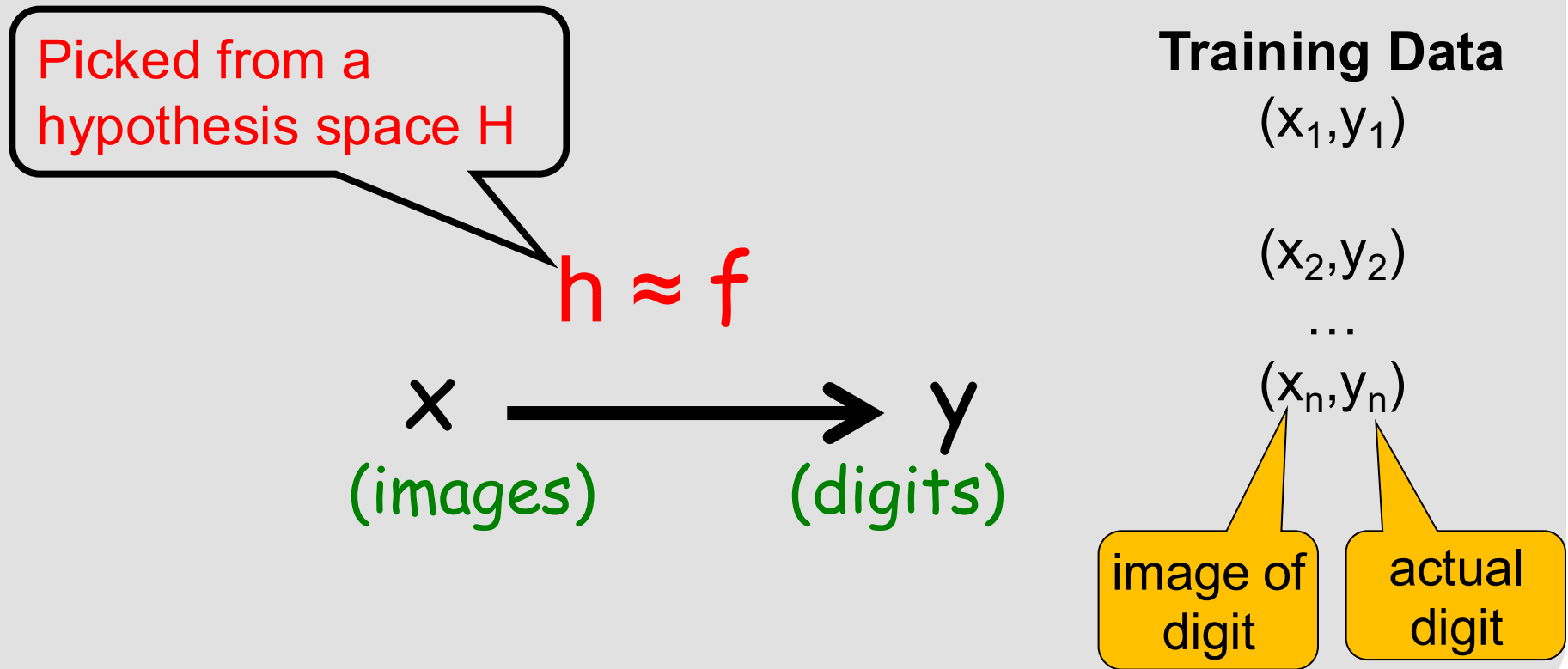
Supervised Learning Example: Handwritten Digit Recognition (I)

- **Classify images into one of the digits: 0, 1, ..., 9**
 - Useful for the post office to automatically detect addresses



Supervised Learning Example: Handwritten Digit Recognition (II)

- Learn a **function f** from examples



Supervised Inductive Learning

- **Setup:**
 - f is the **unknown** target function
 - We are given some sample pairs from it $(x, f(x))$
- **Problem: learn a function hypothesis h**
 - Based on the set of *training* examples
 - Such that $h \approx f$ (h approximates f as best as possible)
 - > h must **generalize** well on unseen examples



Picking the Best Hypothesis h

- **Big Idea 1:**

Pick h from the space H which agrees with f on the training set

– Complete and consistent

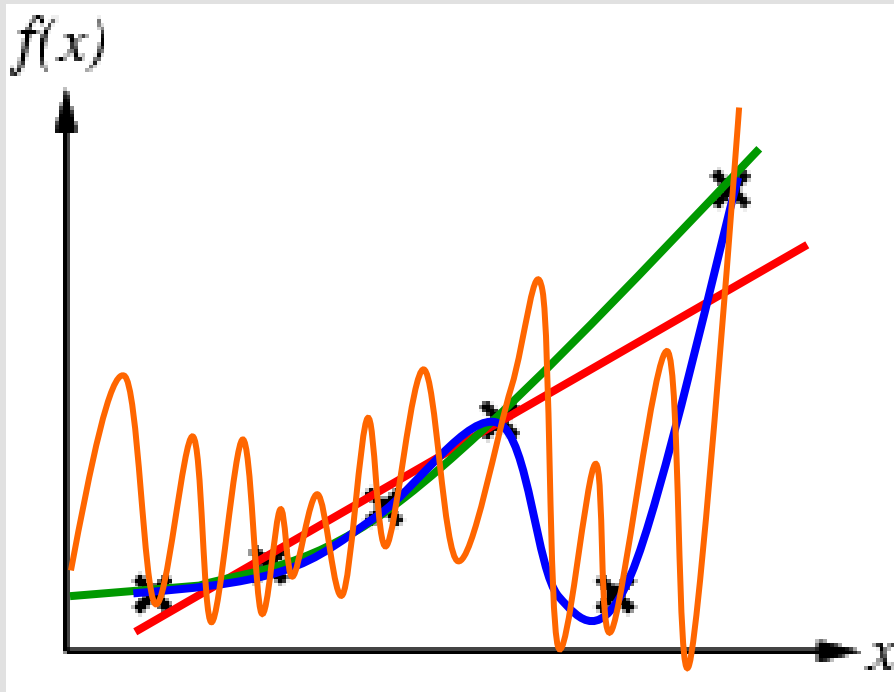
Hypotheses h that are too simple may **underfit** the data – miss relevant relations between features and target outputs



Supervised Learning – Example (I)

Curve fitting (regression):

$h =$ **Straight line?** **Quadratic?** **Cubic?** **Other?**



Training Data

(x_1, y_1)

(x_2, y_2)

...

(x_n, y_n)



Ockham's Razor

- *“Entities must not be multiplied beyond necessity” (“Non sunt multiplicanda entia sine necessitate”)*

William of Ockham (1287–1347)

- **Big Idea 2:** Prefer a simpler hypothesis to complex ones provided both explain the data equally well

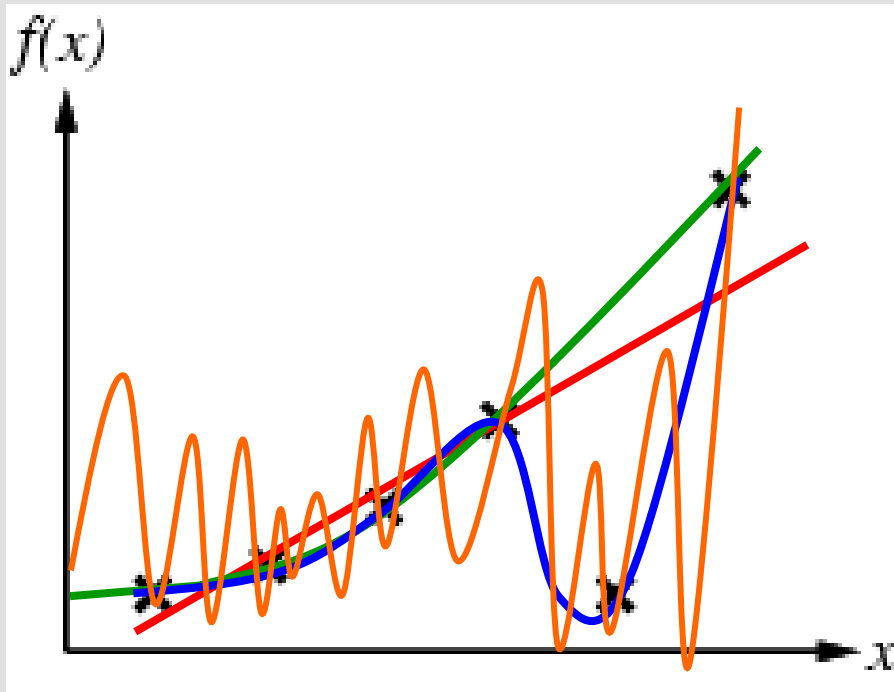
Overly complex hypotheses h tend to **overfit** the data – fit the training data closely, but don't generalize well



Supervised Learning – Example (II)

Curve fitting (regression):

$h =$ **Straight line?** **Quadratic?** **Cubic?** **Other?**



Training Data

(x_1, y_1)

(x_2, y_2)

...

(x_n, y_n)



Mathematical Principles of Learning (I)

Idea 2: To improve
generalizability and
prevent **overfitting**

Bias

Choose the *simplest* hypothesis from
H which is *complete* and *consistent*
with the training data

Idea 1: Should be similar
to **the unknown true**
underlying function



Mathematical Principles of Learning (II)

Best learned hypothesis

Idea 2: penalize complexity – regularization

$$h^* = \operatorname{argmin}_{h \in H} \left\{ \left[\sum_{(x,y) \in D} \operatorname{error}(h(x), y) \right] + \lambda \operatorname{Complexity}(h) \right\}$$

Bias

Idea 1: Error on the training data

Learning is indeed Search/Optimization

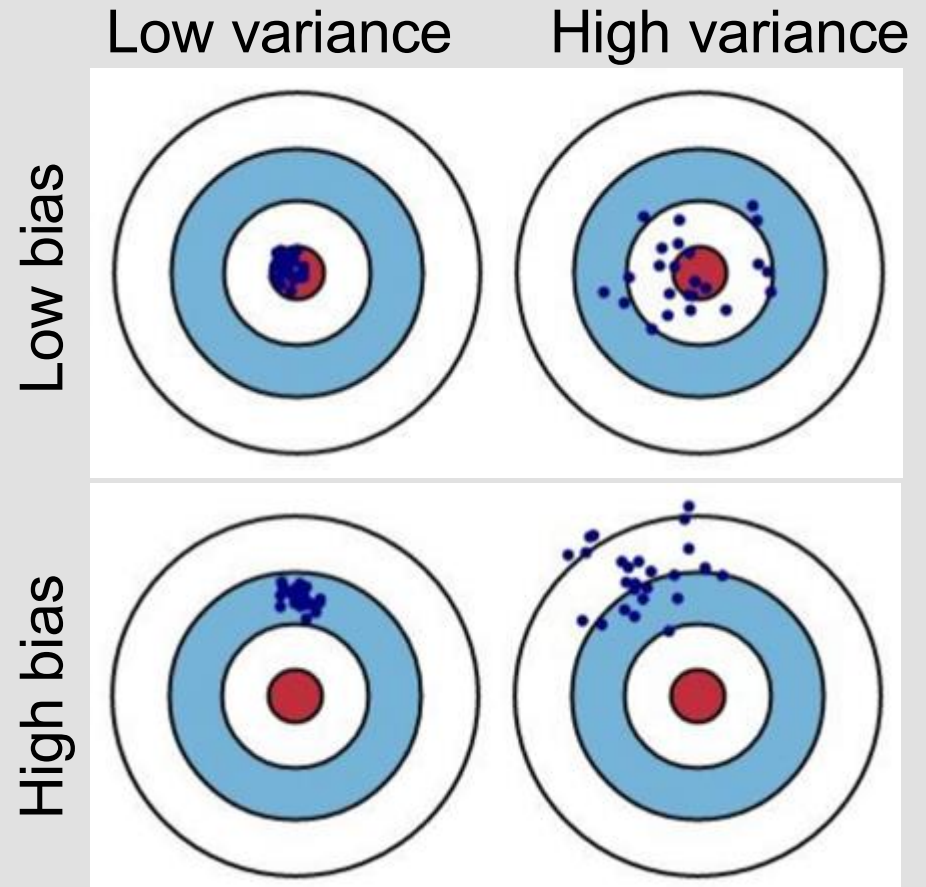


Why may h^* differ from the true function f ?

- **Unrealizability** (due to high bias) – f may not be in H
If H is too simple → underfitting
 - **High variance** – a learning algorithm may be overly sensitive to fluctuations in the training data
 - **Noise** – f may be nondeterministic or noisy
- } overfitting
- **Computational complexity** – when H is complex, it can be intractable to systematically search the whole hypothesis space

Bias and Variance Interaction

- **Bias:**
 - Error due to wrong assumptions about the target function
- **Variance:**
 - Error due to sensitivity to small fluctuations in the training set



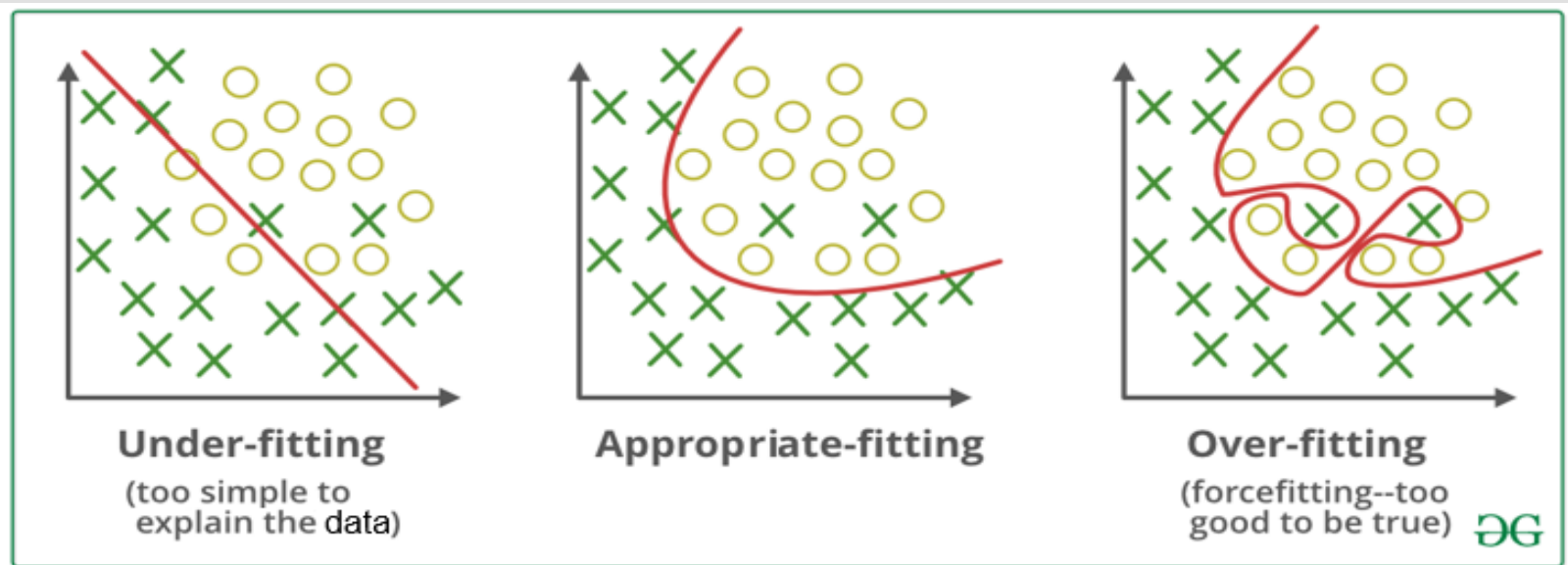
Overfitting and Underfitting

- **Overfitting:**

- A function that corresponds too closely to a given dataset
→ fails to fit new data or predict future observations reliably

- **Underfitting:**

- A statistical model that cannot adequately capture the underlying structure of the data

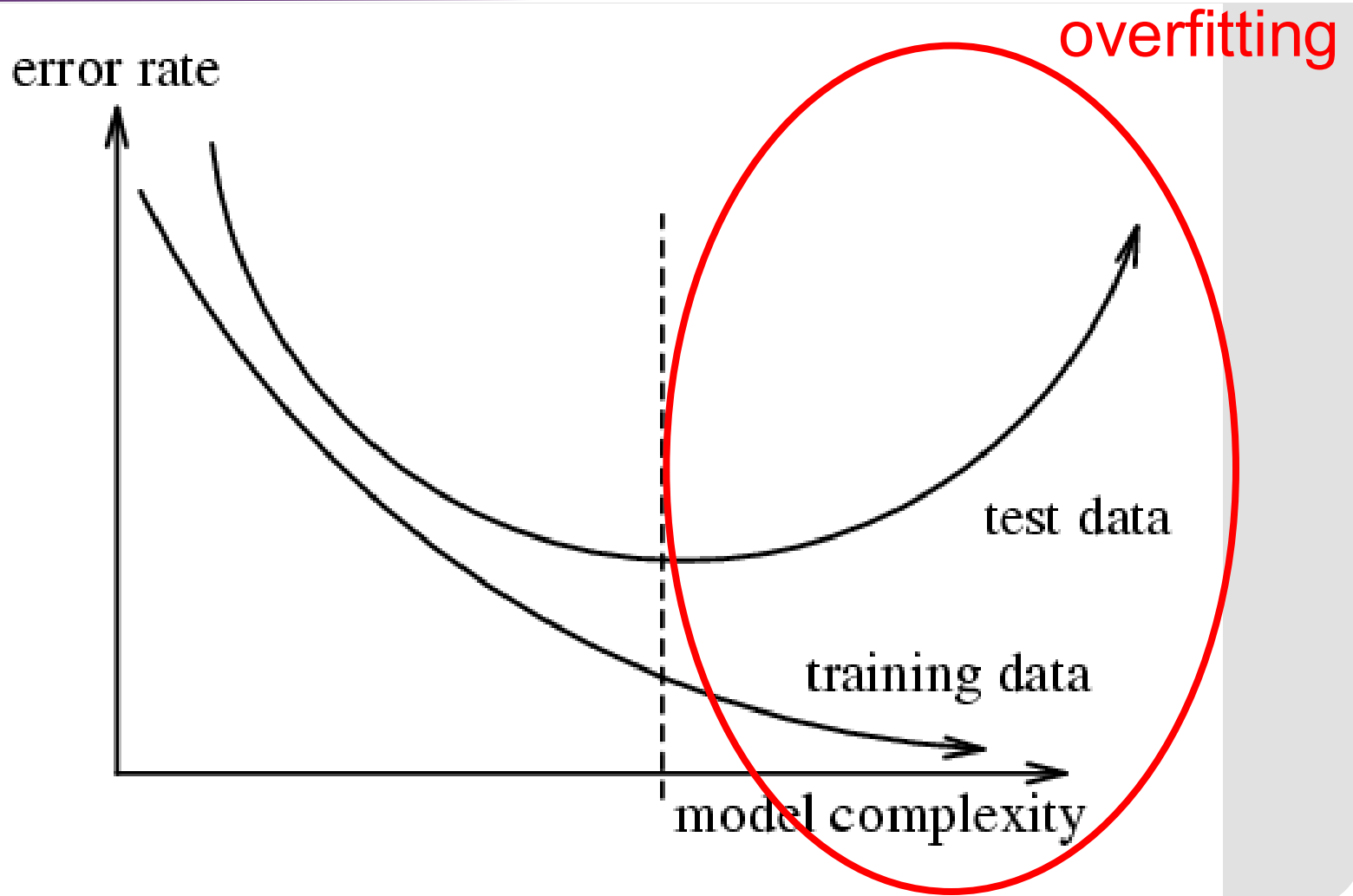


Causes of Overfitting and Underfitting

Overfitting	Underfitting
Noisy data (f may be nondeterministic or noisy)	
High variance	High bias (H is too simple)
Overly complex model (that fits noisy data or outliers)	
Training set is too small	



Model Complexity and Overfitting



How to reduce overfitting?

- Increase the diversity of the training data
- Simplify the model
- Regularization: Add a **complexity** penalty to the performance measure
 - E.g., Complexity = $\mathcal{F}$ (size of the model)
- Fine tune the model on a validation dataset that is different from the training dataset





FIT3080 – Artificial Intelligence

Decision Trees

Decision Trees

- **Classify objects based on the values of their *explanatory attributes***
 - the target classes (*dependent attributes*) are pre-defined
- **Classification is based on a tree structure**
 - each non-leaf node is a *decision node*
 - each leaf node represents a *class*
- **To classify an object**
 - each decision node (starting from the root) compares an attribute of the object with a specific *attribute value* (or range)
 - a path from the root to a leaf node gives the class of the object



Decision Tree Example: Attributes

Day	Outlook	Temperature	Humidity	Wind	Play ball
D1	Sunny	Hot	High	Weak	No

Input x (vector of values for
explanatory attributes)

Output y
(*target / dependent attribute*)



Classification by DT Induction

- **Training data** – records of items that have *explanatory attributes* and a *dependent attribute*
 - **Input x** : Represented by a vector of attribute values
 - **Output y** : The corresponding target value
- **Tree construction algorithm** – builds a decision tree
 - Top-down, recursive divide-and-conquer approach
 - Based on a greedy algorithm
 - > Selects the best option (irrevocable)

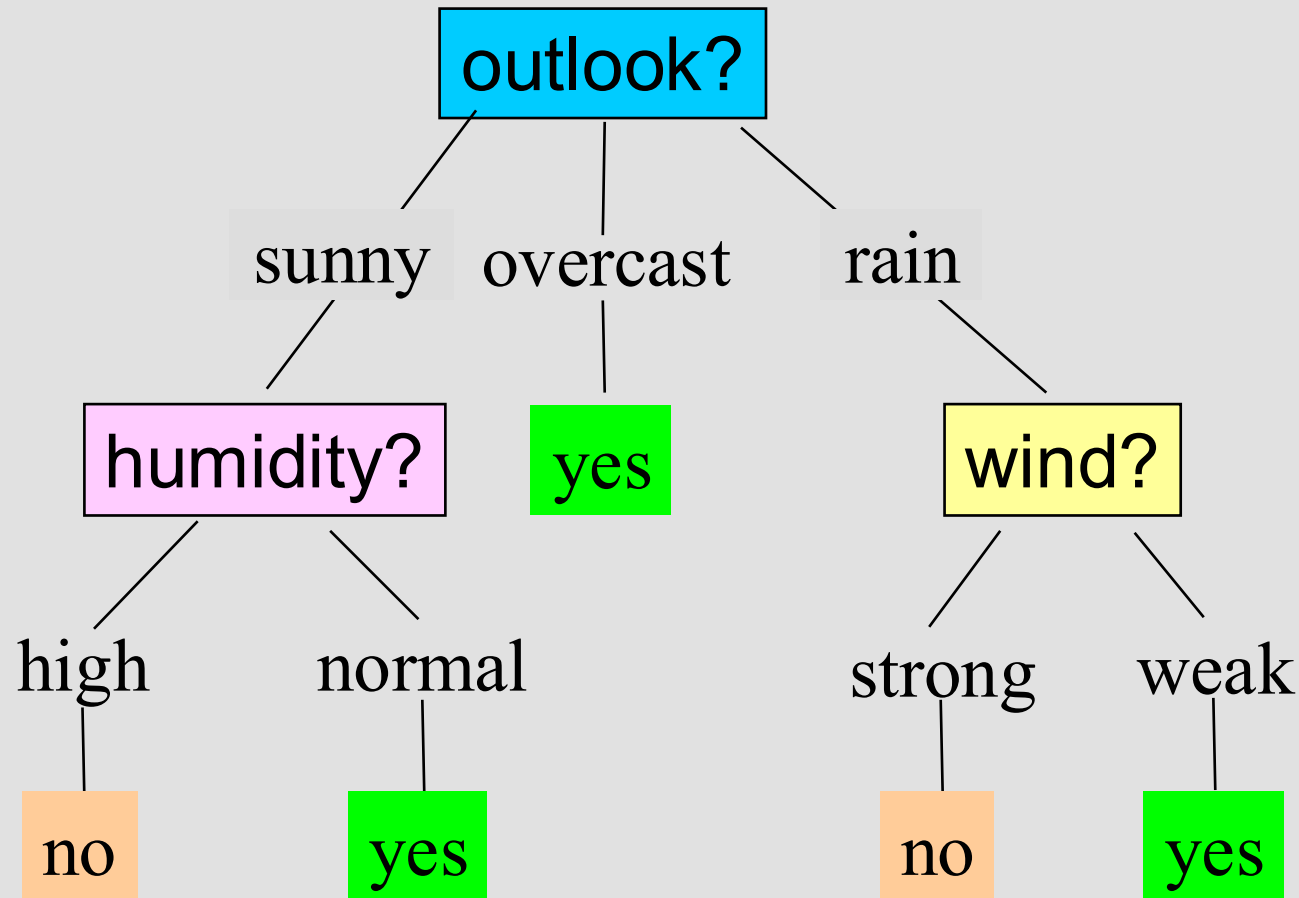


DT Example – Training Dataset

Day	Outlook	Temperature	Humidity	Wind	Play ball
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



DT Example – Learned Model



Decision Tree Learning Algorithm (I)

1. Start with all training examples at the root
 2. Partition examples recursively based on selected attributes
 - attributes are categorical
 - > if continuous-valued, they are broken up into ranges
 - attributes are selected using heuristics or a statistical measure
 - > e.g., *information gain*
 3. Stop partitioning when there is no further gain in partitioning
- Employ majority voting for classifying the leaves**



Choosing an Attribute for Splitting

- **Approach – choose the attribute that best separates training examples into targeted classes**
- **Examples of proposed techniques**
 - ***Information Gain*** [Quinlan, 1975]
 - ***Message Length*** [Wallace and Patrick, 1993]

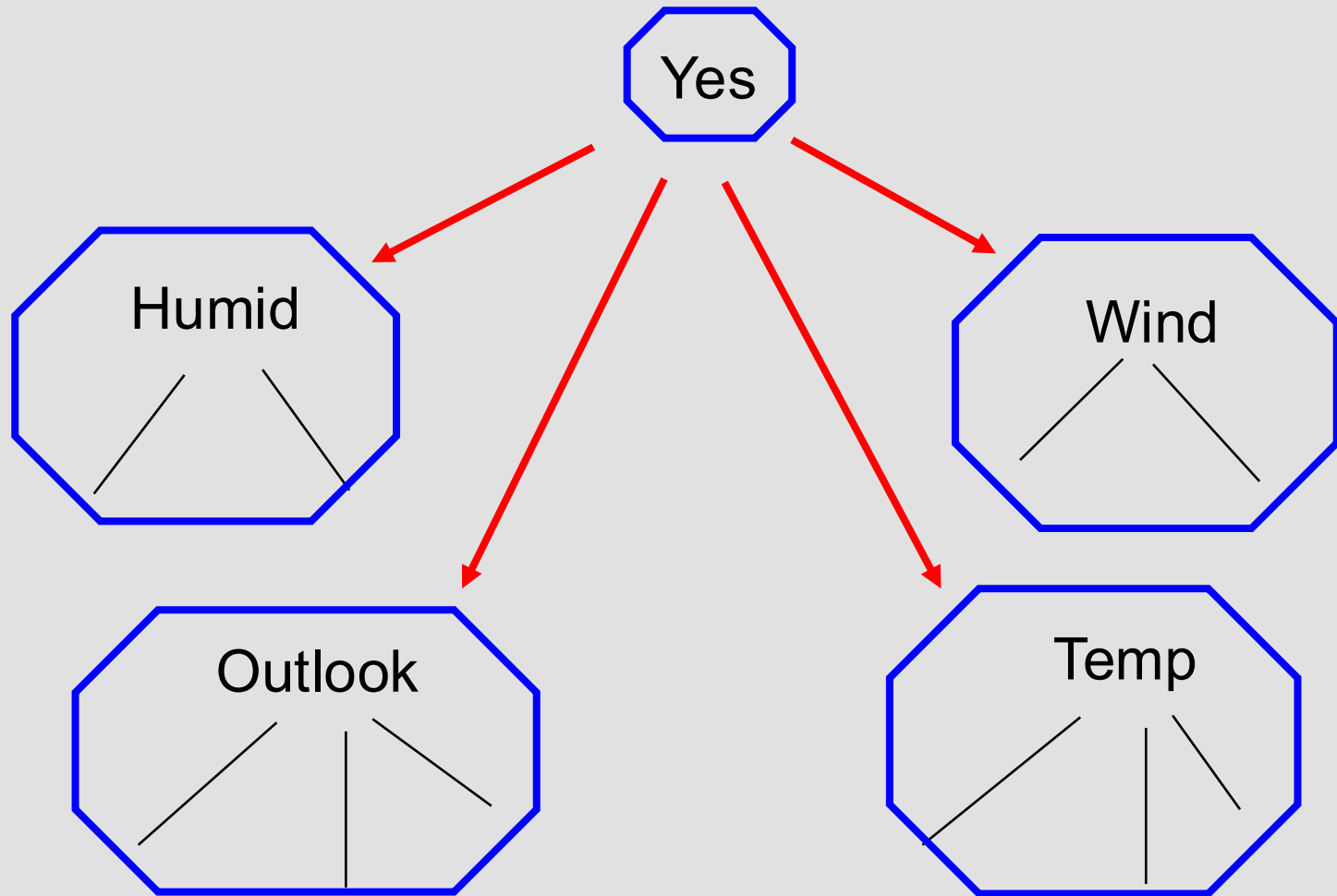


What is the “simplest” Tree? – Example

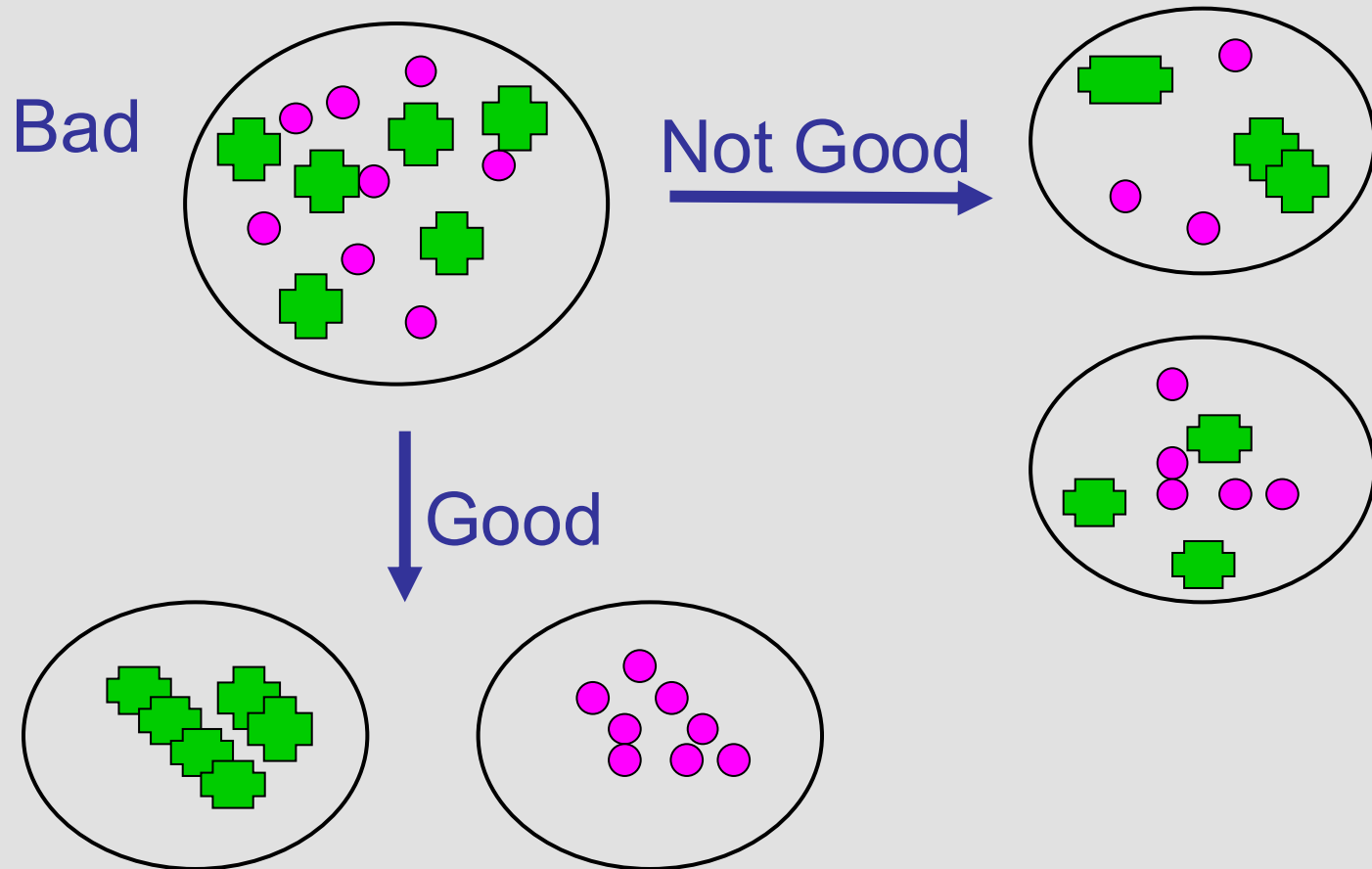
- **Majority class (not a tree)**
 - Always predict “yes”
 - How good it is?
 - > Correct on 9 examples
 - > Wrong on 5 examples
- **Decision stump**
 - A tree with one decision node
 - > Splits on one attribute

Day	Outlook	Temperature	Humidity	Wind	Play ball
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

On which Attribute do we Split?



Disorder is Bad, Homogeneity is Good



Using Information Theory to Quantify Uncertainty – Entropy

- **Entropy** measures the amount of uncertainty in a probability distribution
- Given a discrete random variable on a finite set $X = \{x_1, \dots, x_n\}$, with probability distribution function $\Pr(x) = \Pr(X=x)$, the entropy $H(X)$ of X is defined as

$$H(X) = - \sum_{i=1}^n \Pr(x_i) \log_2 \Pr(x_i)$$

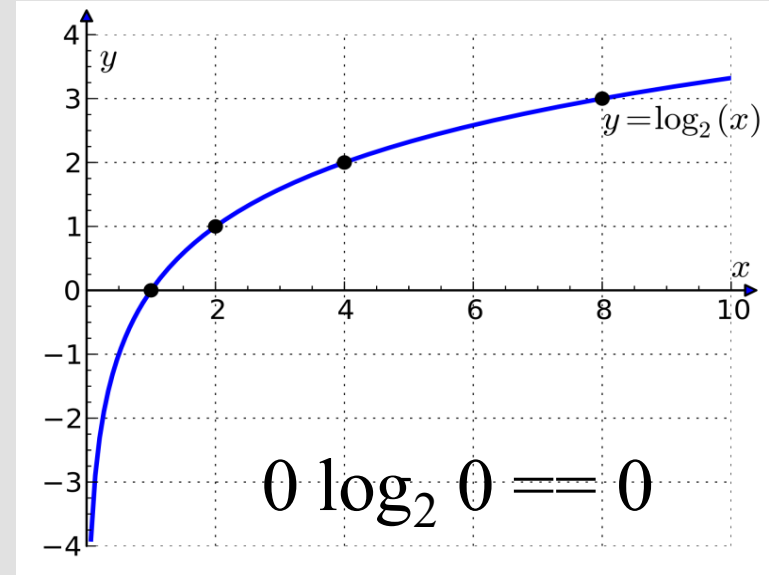
where $0 \leq H(X)$



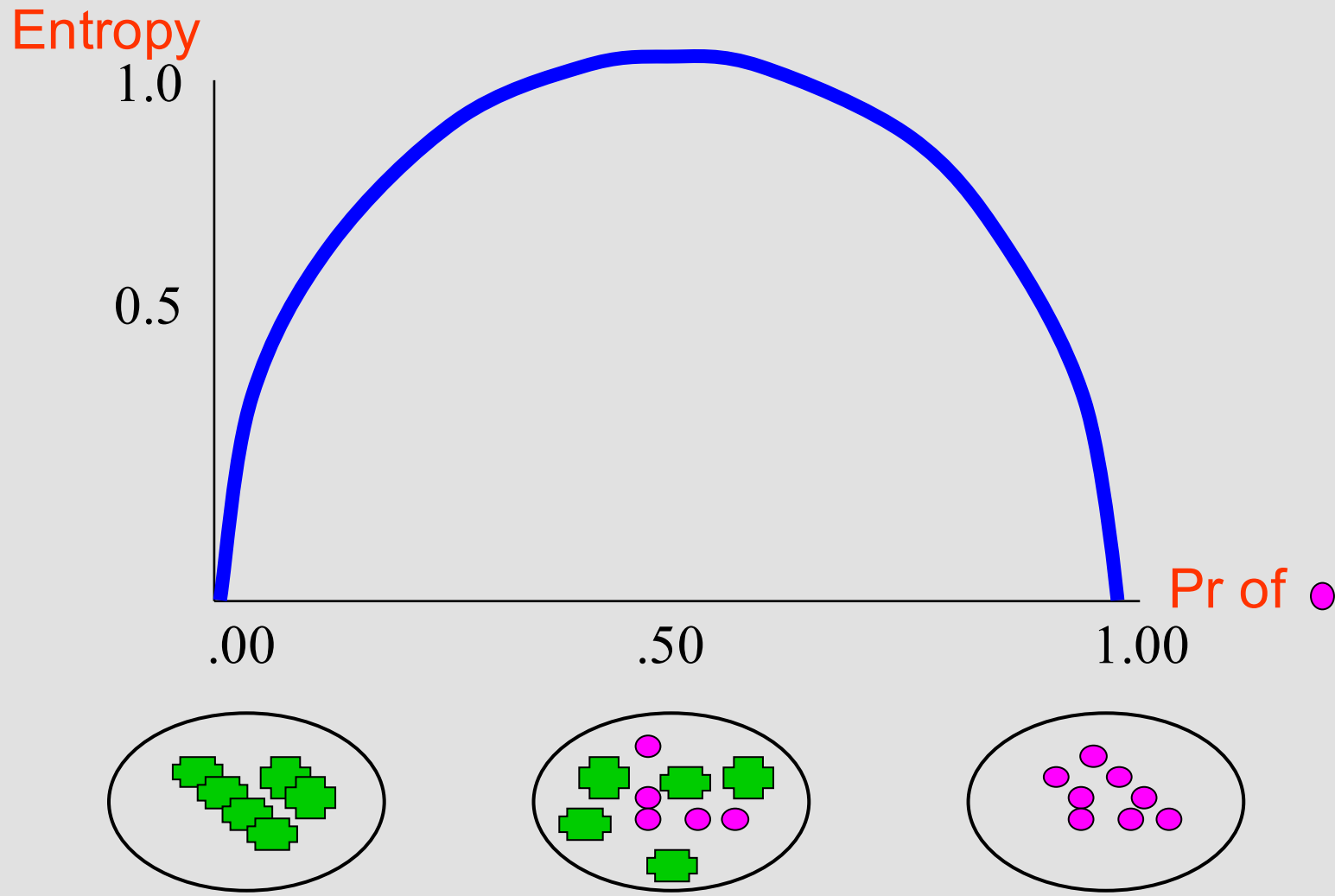
Entropy – Example

$$H(X) = -\sum_{i=1}^n \Pr(x_i) \log_2 \Pr(x_i)$$

- **Suppose there is a random variable S that has value a or b**
 - Let $\Pr(a) = 1$ and $\Pr(b) = 0$
 - Let $\Pr(a) = 0.9$ and $\Pr(b) = 0.1$
 - Let $\Pr(a) = 0.5$ and $\Pr(b) = 0.5$
 - Which probability assignment maximizes $H(S)$?



Entropy



Entropy is Bad, Homogeneity is Good

- **Let S be a set of examples**
 - Labeled positive (P) or negative (N)
- **Entropy(S) = $-\text{Pr}(P) \log_2(\text{Pr}(P)) - \text{Pr}(N) \log_2(\text{Pr}(N))$**
 - **Pr(P)** is the proportion of positive examples
 - **Pr(N)** is the proportion of negative examples
 - $0 \log 0 == 0$
- **Example: S has 9 pos and 5 neg**
 - Entropy([9+, 5-])

$$H(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$



Information Gain (I)

- **Information gain** is a measure of the expected reduction in entropy resulting from splitting along attribute A

$$IG(S, A) = H(S) - \underbrace{\sum_{v \in \text{Values}(A)} \frac{S_v}{S} H(S_v)}$$

Expected value of H from splitting on A

where

- v is a value for A (we sum over all values)
- S_v is a subset of S for which attribute A has value v



Using Information Gain to Learn a DT

- **The same attributes must describe each sample**
- **Attributes are assumed to be categorical (for now)**
- **Select the attribute with the highest Information Gain → largest reduction in entropy**



Information Gain of Splitting on Wind

Values(wind)=weak, strong

$S = [9+, 5-]$

$S_{weak} = [6+, 2-]$

$S_{strong} = [3+, 3-]$

$IG(S, \text{wind})$

$$= H(S) - \sum_{v \in \{\text{weak}, \text{strong}\}} \frac{|S_v|}{|S|} H(S_v)$$

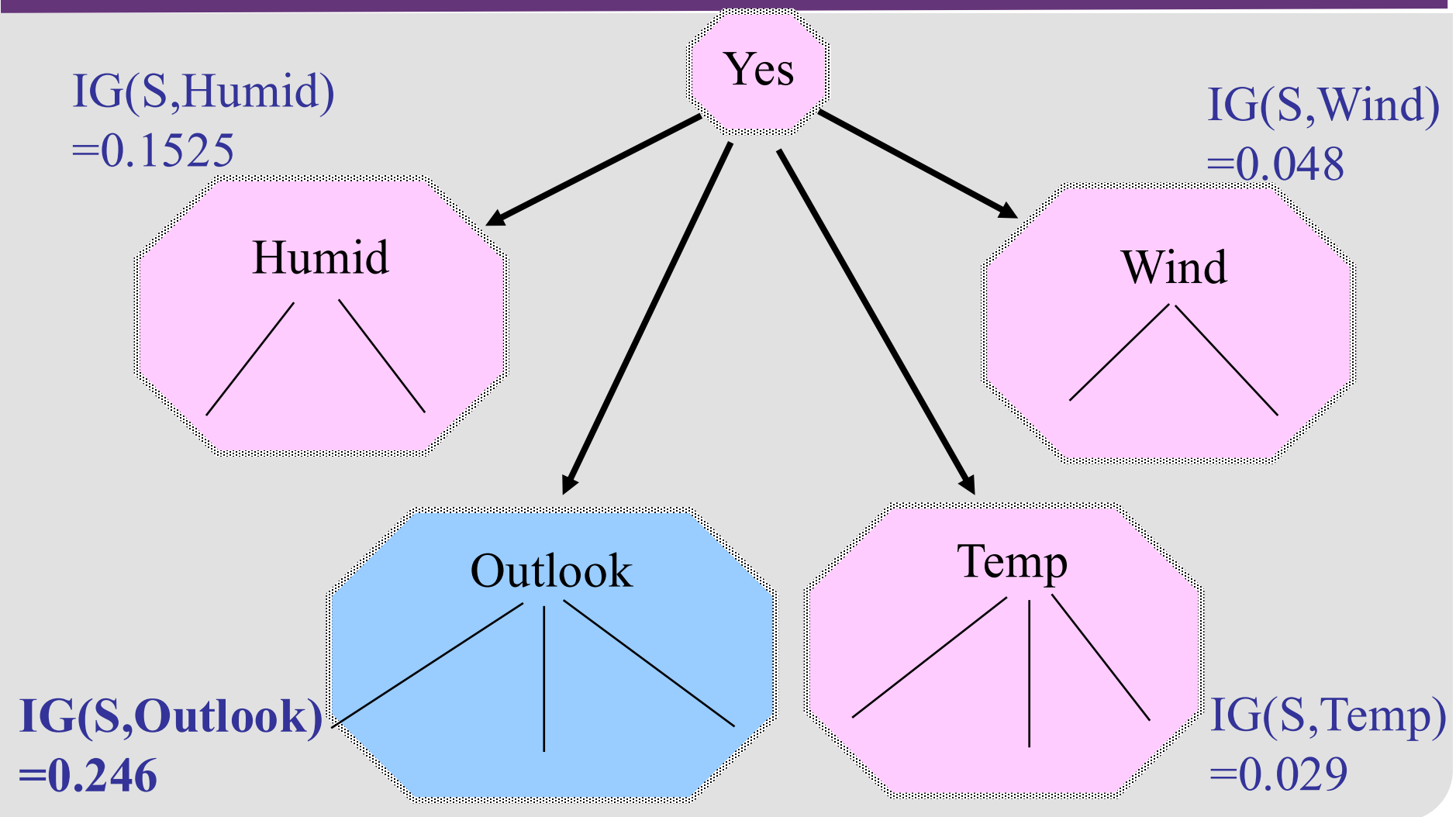
$$= H(S) - \left(\frac{8}{14} H(S_{weak}) + \frac{6}{14} H(S_{strong}) \right)$$

$$= 0.94 - \left(\frac{8}{14} 0.811 + \frac{6}{14} 1.00 \right) = 0.048$$

Day	Wind	Play ball?
d1	weak	no
d2	strong	no
d3	weak	yes
d4	weak	yes
d5	weak	yes
d6	strong	no
d7	strong	yes
d8	weak	no
d9	weak	yes
d10	weak	yes
d11	strong	yes
d12	strong	yes
d13	weak	yes
d14	strong	no

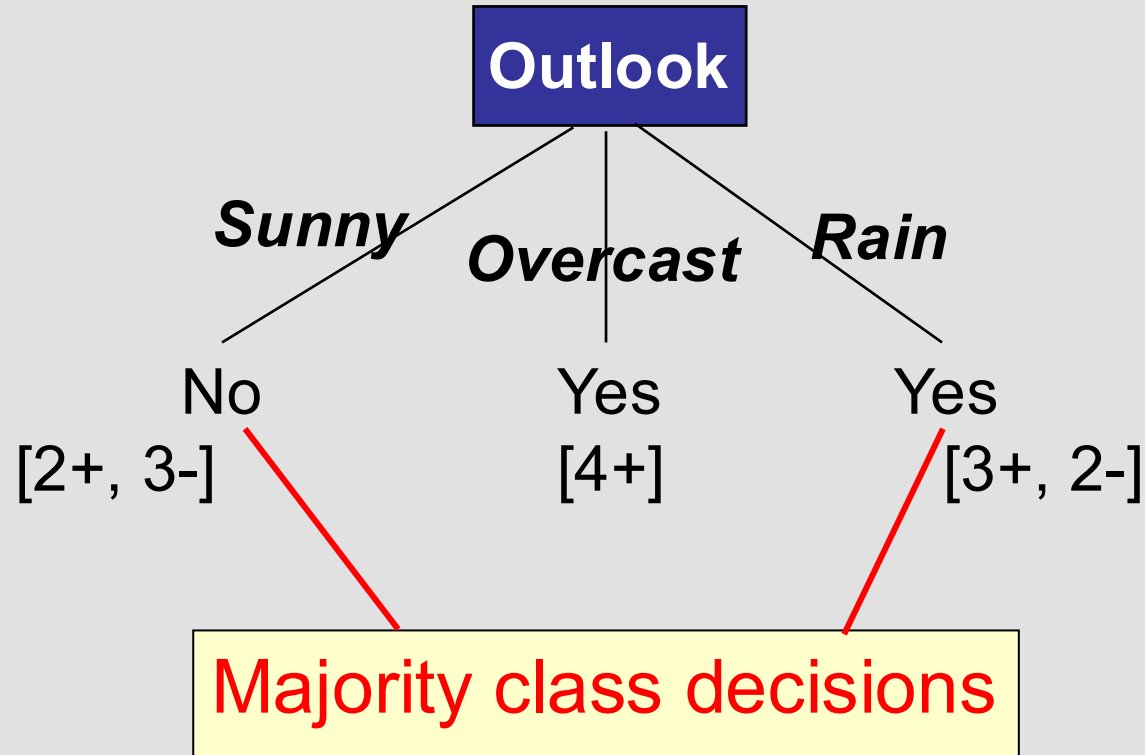


Evaluating Attributes

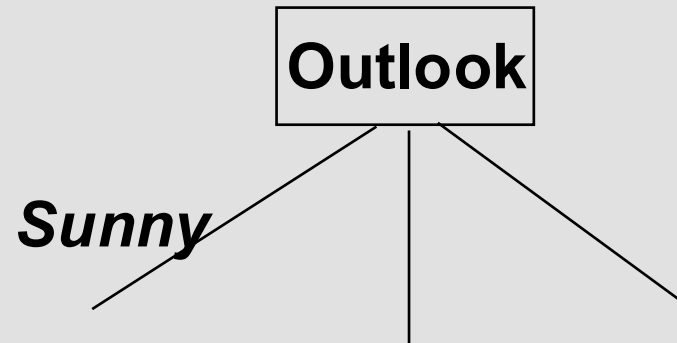


Building a Decision Tree (I)

Good day for playing ball?



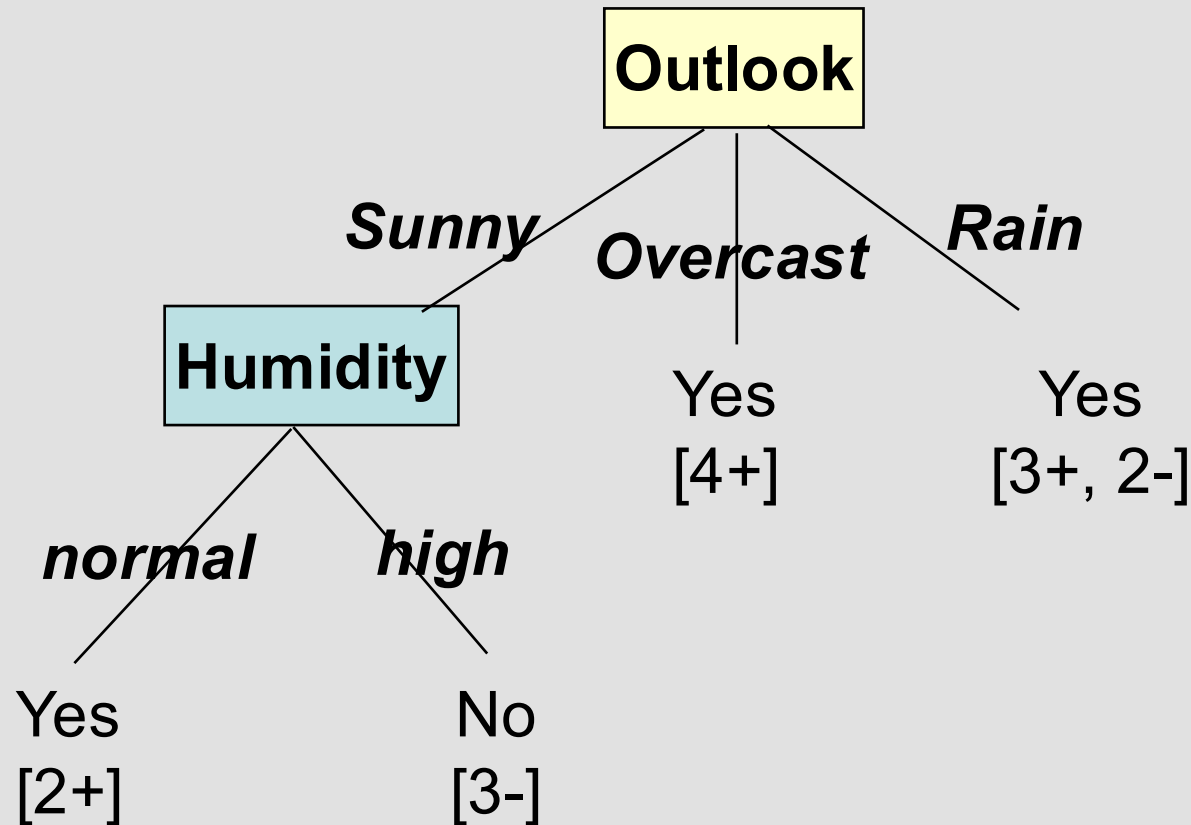
Building a Decision Tree (II)



Day	Temp	Humid	Wind	Play ball?
d1	H	H	weak	no
d2	H	H	strong	no
d8	M	H	weak	no
d9	C	N	weak	yes
d11	M	N	strong	yes



Building a Decision Tree (III)



Decision Tree Learning Algorithm (II)

BuildTree(TrainingData)

- Split(TrainingData)

Split(D)

1. **If** all points in D are of the same class
 Then Return
2. **For each** attribute A
 - a. Evaluate splits on attribute A
3. Use best split to partition D into D1, D2, ...
 - a. Split (D1)
 - b. Split (D2)
 - ...

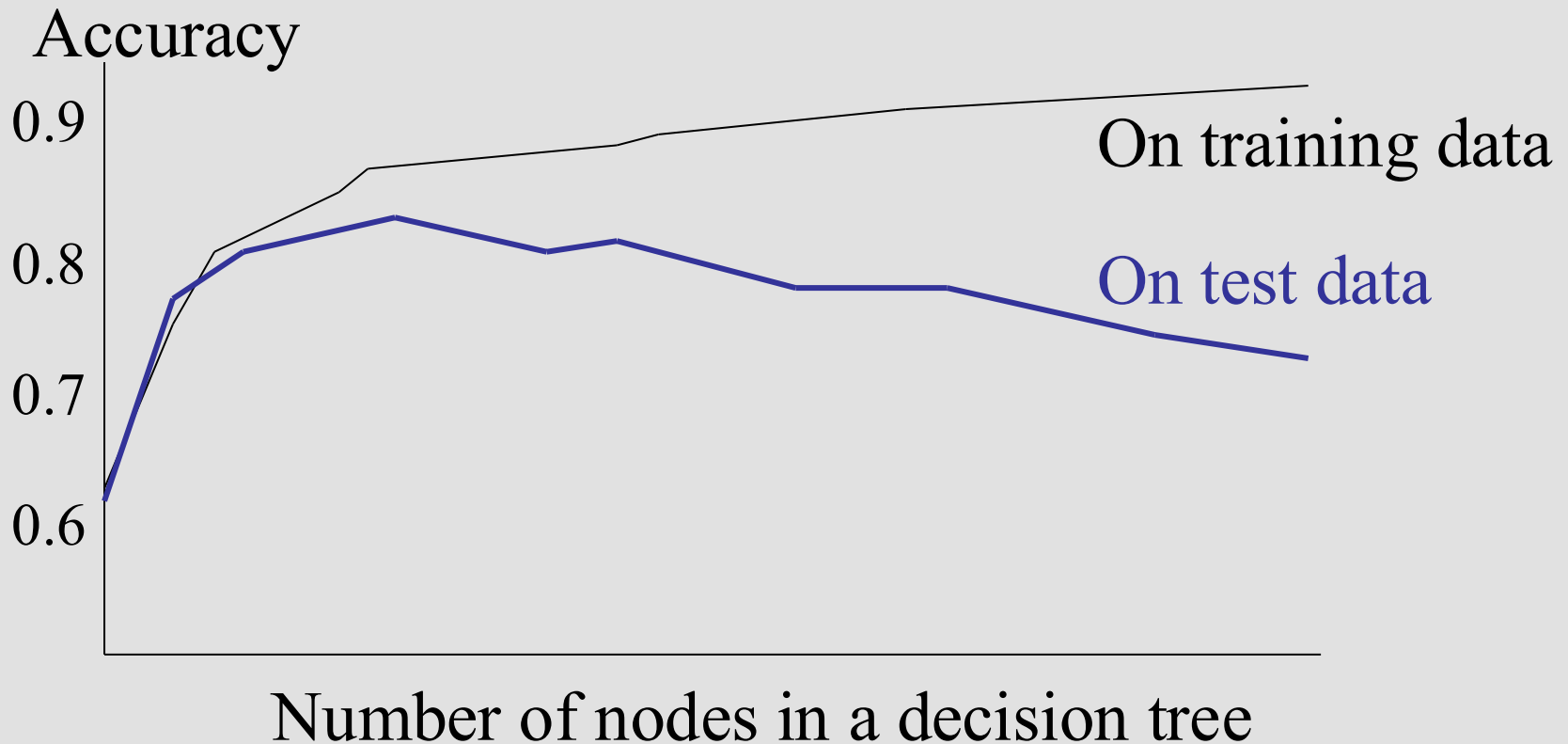
} recursive call



Overfitting and Decision Trees

A generated tree may *overfit* the training data – try to fit noise or outliers

→ reduced accuracy for unseen samples



Reducing Overfitting in Decision Trees

- **How to reduce overfitting:**

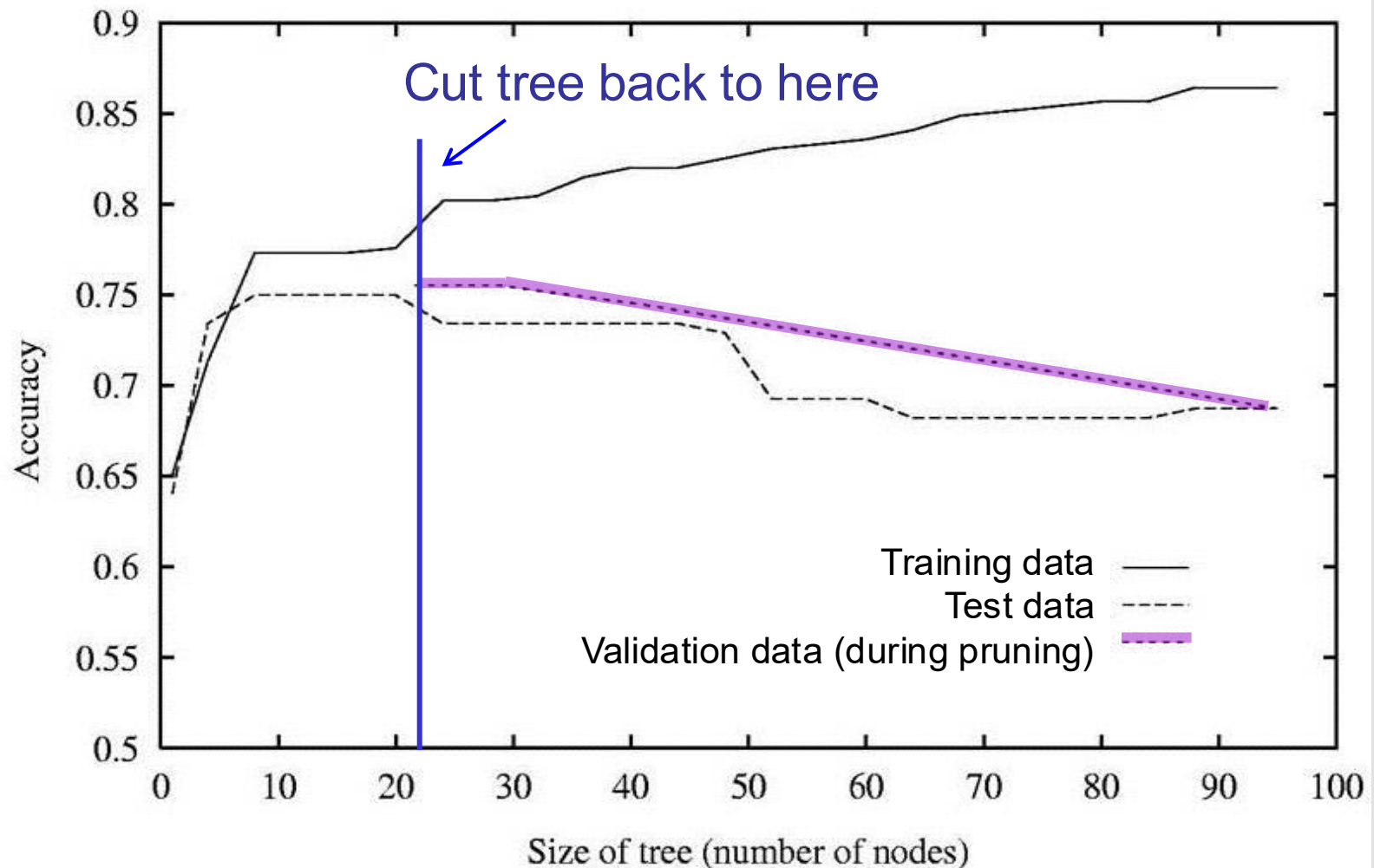
- **Pre-pruning:** Stop growing the tree if the goodness measure falls below a threshold
- **Post-pruning:** Grow the full tree, then prune
 - > Use a **validation dataset**
- **Regularization:** Add a **complexity** penalty to the performance measure
 - > E.g., Complexity = Number of nodes in the tree

- **How to select the best tree?**

- Measure performance on training data
- Measure performance on a separate **validation set**

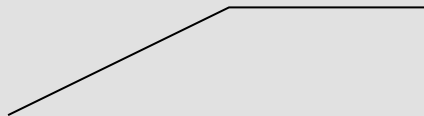


Effect of Pruning after Growing the DT



Enhancements to DT Induction

- **Handle continuous-valued attributes**
 - **Input:** Use threshold to split
 - **Output:** Estimate a linear function at each leaf (e.g., mean)
- **Handle missing attribute values**
 - **Categorical**
 - > use the most common value of the attribute
 - > probabilistic selection according to the distribution of values
 - **Continuous**
 - > use the mean of the attribute values
- **Reduce overfitting**
 - E.g., by post-pruning
- **Attribute construction**
 - create new composite attributes based on existing ones



**done
manually**



Continuous-valued Attributes

Partition the continuous attribute value into intervals

1. **Fit a distribution to the values for attribute A**
 - commonly Normal
2. **Search for a point to split on**
 - perform binary search
 - can split on the same attribute again (lower in the tree)
3. **Calculate the Information Gain obtained from splitting attribute A at that point**



Continuous-valued Attributes – Example

- Humidity has a Normal distribution with mean $\mu=81.6$ and standard deviation $\sigma=10.53$
- When we split on 75 (after outlook=sunny), we obtain

Day	Outlook	Temperature	Humidity	Wind	Play ball
D1	Sunny	Hot	83	Weak	No
D2	Sunny	Hot	94	Strong	No
D8	Sunny	Mild	89	Weak	No
D9	Sunny	Cool	73	Weak	Yes
D11	Sunny	Mild	69	Strong	Yes

$$IG(S, h \leq 75) =$$

$$\begin{aligned}
 & H(S) - \left(\frac{2}{5} H(h \leq 75) + \frac{3}{5} H(h > 75) \right) = \\
 & = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} - \left(0 + 0 \right) = 0.97
 \end{aligned}$$

Reading

- **Supervised learning** – Russell, S. and Norvig, P. (2022), *Artificial Intelligence – A Modern Approach* (4th ed), Prentice Hall
 - Fundamental concepts – Chapter 19.1-19.2
 - Decision trees – Chapter 19.3
 - Evaluating the best hypothesis – Chapter 19.4-19.4.1



Next Lecture Topic

- **Lecture Topic LN9**
 - Machine Learning Part 2 –
 - Regression, Logistic regression and Neural Networks

